

out, they predicted that the magnetic moment should be a little bit bigger than one, though still not quite as big as the measured value, but there is no need to stop there.

“There is one simplification at least. Electrons behave in this respect in exactly the same way as photons; they are both screwy, but in exactly the same way. How they behave, therefore, takes a great deal of imagination to appreciate, because we are going to describe something which is different from anything you know about.”

Richard Feynman (1967)

in the right direction. By the time the effect of four photons being emitted and reabsorbed was allowed for, the predicted value of the magnetic moment was 1.00115965246, while the best experimental measurement was 1.00115965221.

HOLLYWOOD

splitting hairs
In order to equal the accuracy of calculations carried out using QED, the distance between New York and Los Angeles, which is 2,462 miles (3,961 km), would have to be measured to within the thickness of a human hair, which is $\frac{1}{127}$ inch (0.2 mm).

percent. This is the most precise agreement between theory and observation for any experiment carried out. This is powerful evidence that the whole edifice of quantum physics is built on solid foundations, and we can use it with confidence to predict (or explain) the behavior of atoms and molecules.

There is no reason why the electron cannot emit two (or more) photons, one after the other, and reabsorb them. Each extra photon allowed for in the calculations brings the calculated value of the magnetic moment ever closer to the measured value.

Crucially, Feynman was able to *prove* that each time an extra photon is allowed for in the calculation, the “correction” is a little bit smaller, but always

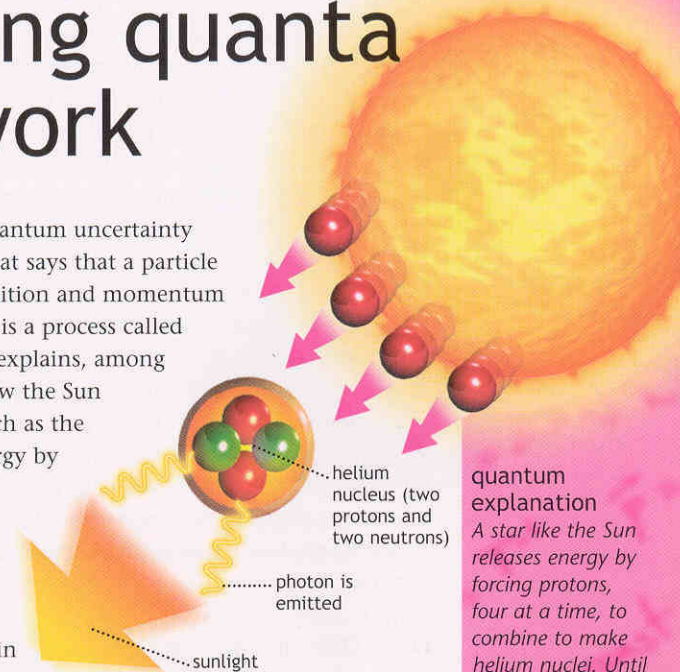
Theory and experiment agreed to an accuracy of one part in 10 decimal places, or 0.00000001



putting quanta to work

One result of quantum uncertainty (the principle that says that a particle cannot have position and momentum simultaneously) is a process called tunneling. This explains, among other things, how the Sun shines. Stars, such as the Sun, release energy by the process of nuclear fusion.

In order for nuclei to fuse, two positively charged nuclei (in the simplest case, two nuclei of hydrogen, each consisting of one proton) must come together. But according to classical electromagnetic theory, this is impossible because the two positively charged particles repel each other and cannot fuse. Quantum physics provided an explanation of how fusion might be possible. Due to quantum uncertainty, if two protons are very close together, it is not certain if they are touching or not, and they may or may not fuse. Another way to think about this is in terms of waves. If the quantum entities can approach each other so closely that their wave functions overlap, they can be pulled together by the interacting wave functions. This is called “tunneling” because the two particles come up against the electric barrier of classical physics that exists between



quantum explanation
A star like the Sun releases energy by forcing protons, four at a time, to combine to make helium nuclei. Until quantum theory was applied to the problem, this was a great puzzle, because the temperature inside the Sun is not high enough to do this without the help of quantum uncertainty.